Primary Adrenal Insufficiency Due to Bilateral Adrenal Hematomas in a Patient with Primary Antiphospholipid Syndrome

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To the Editor: In 2016, a 35-year-old Chinese man was admitted to our hospital who had experienced multiple thromboses for 9 years, severe fatigue, and oliguria for a total of 3 weeks. The patient had been diagnosed with pulmonary embolism and multiple deep vein thromboses at a local hospital 9 years ago. The patient was prescribed warfarin, and the international normalized ratio (INR) was maintained between 2.42 and 3.18. While laboratory findings revealed a prolonged case (77.8 s) of activated partial thromboplastin time (APTT) at that time, no further investigations were carried out and anticoagulation therapy for the patient was terminated after 3 months.

In March of 2016, the patient was admitted to a local hospital for acute cholelithiasis and cholecystitis. Laboratory tests revealed significantly prolonged APTT (74.5 s). Fatigue and oliguria were observed in the patient following laparoscopic cholecystectomy and amoxicillin administration. Urine output was observed to increase to 1000 ml/d following the withdrawal of antibiotics, while fatigue was exacerbated with a low-grade fever and a loss of appetite.

This patient was transferred to our hospital with a fever of 39.5°C and low blood pressure (BP, 86/48 mmHg, 1 mmHg = 0.133 kPa). In addition, urinary tract infection (UTI), acute kidney injury (Creatinine [Cr] between 405 and 453 μmol/L), hyponatremia (132 mmol/L), hyperkalemia (5.8 mmol/L), prolonged APTT (105.1 s, normal range 22.7–31.8 s), and increased levels of D-dimer (6.96 mg/L) were detected. The patient's platelets count was low (85 × 10^9/L, normal range 100–300 × 10^9/L) at first, but then spontaneously normalized 2 days later and was maintained at a normal level afterward. Inflammation markers including white blood cells and procalcitonin were measured as normal. Blood cultures and T-SPOT.TB test were also both negative, and no evidence of sepsis was detected. With supportive therapy, meropenem for the UTI, and plasma transfusion (2 mg dexamethasone was given routinely before blood product transfusion), the patient's BP rose to 100/60 mmHg, and serum Cr levels decreased to 146 μmol/L. Meanwhile, the patient's fatigue was slightly alleviated.

He was then admitted to the hematology ward. Physical examination revealed hyperpigmentation of the skin and mucosa. The patient denied photosensitivity, oral or genital ulceration, arthralgia or Reynaud's phenomenon. Both his medical and family histories were not remarkable. Antiphospholipid syndrome (APS) was suspected due to the prolonged APTT at both 0 and 2 h following incubation with normal serum and the positive antiphospholipid antibodies results (lupus anticoagulant 2.99, anticardiolipin IgG 32 U/ml, and anti-β2-glycoprotein I antibodies 155 RU/ml). To obtain a differential diagnosis of secondary APS syndrome, antinuclear antibodies (ANA), anti-extractable nuclear antigen antibodies, serum complements levels, Coombs test, and urine protein for 24 h was examined to exclude autoimmune and rheumatic diseases, such as systemic lupus erythematosus. With the exception of ANA, which was only marginally positive (1:80), no positive results were found. In addition, no evidence was identified regarding other inherited and acquired thrombophilias, such as protein C or S deficiency, paroxysmal nocturnal hemoglobinuria, or hyperhomocysteinemia. The blood vessels ultrasound and nuclear medicine ventilation-perfusion (V/Q) scan revealed thromboses in the left popliteal vein and pulmonary arteries. A computed tomography scan revealed high-density masses in bilateral adrenal glands that were highly suggestive of bilateral adrenal gland hematomas [Figure 1]. The clinical picture of abnormal electrolytes results and hyperpigmentation of the skin and mucosa elicited investigation into Addison's disease due to bilateral adrenal hemorrhage, which was then confirmed by the detection of low plasma cortisol levels (3.8–11.0 μg/L) and high adrenocorticotropin levels (524–888 pg/ml), leaving the function of adrenal reticular zone and adrenal medulla normal.

Hydrocortisone replacement therapy was immediately started. Because of the thrombosis, the patient was given subcutaneous low-molecular-weight heparin, and then switched to a long-term

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Figure 1: Sagittal computed tomography reconstruction of adrenal glands showing bilateral adrenal hemorrhage. Arrow: Slightly high density masses indicating bilateral hematomas in adrenal glands.

warfarin therapy with a target INR of 2–3. APTT of these patients decreased to 64.8 s on discharge day. Throughout the 11 months of follow-up, the patient was in good condition, though hydrocortisone treatment had to be maintained. No remarkable side effects were observed, such as hemorrhage.

The case described here was that of a young man with multiple thromboses present in both arterious and venous systems. Despite the hypercoagulative state, APTT was found to be prolonged. In addition, this patient developed acute kidney injury, persistent fatigue, and hypotension during the hospitalization, but responded well to anticoagulation and hormone replacement therapy.

The presence of elevated D-dimer levels and multiple thromboses in the absence of known risk factors, including immobilization, drugs, or trauma, in addition to the correctable prolonged APTT and positive antiphospholipid antibody results, led to the diagnosis of APS. Considering that APTT had been prolonged for 9 years ago, antiphospholipid antibodies could not be transiently positive.[1] In addition, no proof of secondary causes, including rheumatic diseases, medication, or malignancies were identified. Therefore, primary APS was diagnosed.

Although APS patients have a tendency to be in a hypercoagulative state, a review of 86 cases of adrenal involvement in APS patients summarized that adrenal hemorrhage was found to be more common than infarction (40% vs. 10%) and that the majority of hemorrhage cases were bilateral.[2]

Two pathogenic mechanisms of adrenal hemorrhage in APS have been proposed. The first mechanism claims that the special vascularization of adrenal glands is responsible for the imbalance between a rich arterial supply and limited venous drainage. It is the abrupt transition of artery-capillary plexus promotes venous thrombosis and subsequent postinfarction hemorrhage. The second proposed mechanism suggests that a large amount of membrane lysobisphosphatidic acid contained in the zona fasciculata, a target of antiphospholipid antibodies, could play a role. These antibodies promote apoptosis and the release of lysosomal proteinases, which activates endothelial cells and leads to the onset of microthrombosis.[3] The symmetrical anatomical and histological structures help to explain the bilateral pattern associated with adrenal hemorrhage. APS causes hemorrhagic infarction of the adrenal glands and is typically associated with stress factors, including surgery, infection, or hemorrhagic diathesis associated with anticoagulation therapy.[2] The formation of thrombi as a result of insufficient anticoagulation could also result in adrenal insufficiency.[4]

Anticoagulation therapy remains the standard therapy despite its association with the formation of hematomas. The target INR range for this therapy remains controversial. In a single series study, it was found that 80% of physicians chose to maintain INR between 3 and 3.5 in APS patients with adrenal hemorrhage-infarction in an effort to reduce thrombotic recurrences. However, other reports suggest that an INR below 3 should be maintained. With intensive anticoagulation treatment, the bleeding rate has been documented to be approximately 3.9% per patient per year.[5] In the absence of recommendations, we prefer to take a more conservative approach in an effort to prevent further hemorrhage and we set the INR target between 2 and 3.

The majority of patients with adrenal involvement in APS have been shown to develop an irreversible cortisol deficiency and atrophy of the adrenal glands. However, few patients have demonstrated an adequate cortisol response to Synacthen during the follow-up period.[3] Hypothalamus-pituitaria-adrenal axis function monitoring was carried out for the patient in the case of recovery.

Declaration of patient consent

The authors certify that they have obtained all appropriate patient consent forms. In the form, the patient(s) has/have given his/her/their consent for his/her/their images and other clinical information to be reported in the journal. The patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.

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Conflicts of interest

There are no conflicts of interest.

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